EV MARKET INTELLIGENCE DASHBOARD

Project Documentation

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Project Domain: Electric Vehicle Market Intelligence & Business Analytics

1. INTRODUCTION

1.1 Project Overview

The Electric Vehicle (EV) industry is one of the fastest-growing sectors in the global automotive market. Its expansion is driven by environmental awareness, government incentives, battery-technology advancements, and demand for sustainable transportation. This project develops an EV Market Intelligence dashboard that converts vehicle, battery, sales, pricing, manufacturing, driving-range, and customer-rating data into meaningful business insights.

1.2 Purpose

The purpose is to provide an interactive analytical solution for understanding EV market performance. The dashboard helps stakeholders evaluate annual sales, manufacturers, vehicle categories, battery technologies, driving range, customer ratings, and manufacturing distribution to support strategic planning, product positioning, research, and sustainable-mobility decisions.

2. IDEATION PHASE

2.1 Problem Statement

The EV market generates large volumes of information across vehicle specifications, sales, prices, battery technologies, driving range, manufacturing countries, and customer ratings. Raw data makes it difficult to compare manufacturers, identify high-growth segments, and understand technology trends. The project therefore aims to create an interactive dashboard that converts these data points into clear visual insights.

2.2 Empathy Map Canvas

The project considers three users: Emily Johnson, an EV Market Strategy Director interested in sales, manufacturers, customer satisfaction, battery adoption, range, and high-growth segments; David Chen, an Automotive Business Intelligence Manager focused on market share, manufacturer performance, battery distribution, production geography, and ratings; and Dr. Sarah Williams, a Sustainable Mobility Research Analyst studying battery innovation, range, manufacturing capability, customer adoption, and sustainability.

2.3 Brainstorming

Proposed dashboard ideas include KPI cards, sales by vehicle category, manufacturer comparison, market-share analysis, battery-technology distribution, driving-range comparison, customer-rating analysis, manufacturing-country analysis, and interactive filters for manufacturer, category, battery technology, and country.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The expected journey is: open dashboard → review KPIs → select filters → compare categories and manufacturers → explore battery and range trends → examine customer ratings and manufacturing geography → derive business or research insights.

3.2 Solution Requirement

Functional requirements include data loading, KPI display, sales comparison, manufacturer analysis, battery analysis, range analysis, customer-rating analysis, manufacturing distribution, and interactive filtering. Non-functional requirements include usability, clear visualization, responsiveness, data consistency, and intuitive navigation.

3.3 Data Flow Diagram

EV Market Dataset → Data Preparation/Cleaning → Tableau Data Connection → Calculations and Aggregations → Visualizations → Interactive Dashboard → User Insights and Decisions.

3.4 Technology Stack

Primary technology: Tableau. Supporting components include CSV/tabular data, data preparation and cleaning, calculated fields, dashboard filters, KPI cards, charts, and interactive visualization techniques.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The dashboard addresses the need to convert fragmented EV market information into an interactive analytical interface. Sales, manufacturers, battery technologies, categories, range, ratings, and manufacturing geography can be explored together.

4.2 Proposed Solution

The proposed solution is a Tableau-based EV Market Intelligence Dashboard combining multiple visualizations into one interactive experience. Users can begin with summary KPIs and then explore detailed market views.

4.3 Solution Architecture

Data Layer → Preparation Layer → Visualization Layer → Presentation Layer. The data layer contains EV market attributes; preparation handles cleaning and analytical fields; visualization contains Tableau worksheets; presentation provides the final interactive dashboard.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Stages: understand the problem and personas; inspect and prepare data; define KPIs; create Tableau visualizations; combine them into a dashboard; add filters and formatting; test calculations and performance; capture final outputs; document and publish the project.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing should verify correct values, filter behavior, chart interactions, labels, legends, and dashboard loading. Functional checks should cover sales, manufacturers, battery categories, range, ratings, and country/category filters. Performance checks should cover loading time, filtering responsiveness, and efficient calculations.

7. RESULTS

7.1 Output Screenshots

Insert screenshots of the completed Tableau dashboard and major visualizations here, including executive KPIs, sales/category analysis, manufacturer analysis, battery technology analysis, range analysis, customer ratings, and manufacturing distribution. Each screenshot should have a short caption.

8. ADVANTAGES & DISADVANTAGES

Advantages

Interactive analysis; comparison across manufacturers and categories; combined sales, battery, range, ratings, and manufacturing views; easier trend identification; support for data-driven decisions; reduced manual analysis.

Disadvantages

Results depend on source-data quality; incomplete historical data can limit trends; very large datasets may require optimization; customer ratings can be subjective; dashboard insights do not by themselves establish causation.

9. CONCLUSION

Conclusion

The EV Market Intelligence project provides a structured way to analyze a rapidly evolving electric-vehicle market through interactive visualization. By combining manufacturer performance, categories, sales, battery technologies, driving range, customer ratings, and manufacturing distribution, the dashboard helps users move from raw information to actionable insights.

10. FUTURE SCOPE

Future Scope

Future enhancements may include forecasting, regularly refreshed market data, charging-infrastructure analysis, regional adoption, total cost of ownership, sustainability metrics, predictive demand models, advanced competitive benchmarking, automated data pipelines, machine learning, and mobile-friendly deployment.

11. APPENDIX

Source Code (if any)

Add supporting preprocessing scripts or other project code here if used.

Dataset Link

Insert the final dataset/repository link here.

GitHub & Project Demo Link

Insert the actual GitHub repository and Tableau/project demo URLs here.